



Information Technology Law: The Law and Society (5th edn)

Andrew Murray

p. 37 3. Net neutrality

Andrew Murray

<https://doi.org/10.1093/he/9780192893529.003.0003>

Published in print: 08 June 2023

Published online: August 2023

Abstract

This chapter examines the nature of network neutrality or net neutrality. This is a highly prized design feature of the network that all data packets shall be treated equally. It was the nature of the early network that this design feature was protected as established by visionaries such as Vint Cerf and Bob Kahn. Tim Berners-Lee the creator of the World Wide Web has written at length as to how net neutrality allowed him to create the web without asking permission from anyone. However, modern network design challenges this ‘first amendment of the internet’ and recent developments in particular in the United States have led to debate on whether net neutrality should be legally enshrined.

Keywords: net neutrality, FCC open internet order, throttling, zero rating, access tiering, European regulation of net neutrality

Network neutrality, or more commonly net neutrality, is highly prized by many internet pioneers including Professor Sir Tim Berners-Lee, creator of the World Wide Web.¹

Highlight Net neutrality defined

Net neutrality is the principle that data packets on the internet should be moved impartially, without regard to content, destination, or source. It is sometimes referred to as ‘The First Amendment of the Internet’.

This previously widely accepted concept has been subject to intense debate and review in recent years and remains at the heart of some of the most intense battles over internet regulation.²

p. 38

Modern routers allow network carriers to prioritize certain traffic over others. Network providers argue this is a positive development as it allows them to prioritize traffic with a low latency threshold such as VoIP and streaming media over traffic with a higher latency threshold such as web browsing or a file download.³ Network providers argue that in this way they can make a more efficient use of the limited resources available to them with everyone receiving the best service possible.⁴ Critics argue that it also allows them to discriminate against certain applications or data types.⁵ There are concerns ↵ that service providers may seek to deteriorate the quality of service of applications or services which compete with their products.⁶ In addition, they may place restrictively low upload speeds on your service making it difficult to use file or video sharing services such as YouTube or BitTorrent, or as was discovered by Professor Tim Wu when he surveyed ISPs' service contracts, you may simply be contractually barred from certain activities by your ISP.⁷

3.1 Challenges to net neutrality

As noted there may be good reason for Internet Service Providers to carry out network traffic management. If, as is the case in North America, and increasingly the case in Europe, two service providers (Netflix and YouTube) are consuming between 29–35 per cent of the available network capacity then it seems sensible that ISPs may want to manage that content to ensure that all users of their network get a reasonable experience online. It certainly doesn't seem fair that because my neighbour is streaming *The Sandman* in 4K on Netflix I should find that my browsing a price comparison site is affected. Returning to our discussion in chapter one, although bits are non-rivalrous there is only so much capacity in the telecommunications network to carry them. My neighbour's use of the network is rivalrous to my use.

As UHD video streaming has become commonplace the strain on the network has become pronounced. If we are to enjoy high latency services such as VoIP/Videoconferencing and streaming video we are faced with two choices: massive investment in improving network capacity, or network management. As the investment programme takes time to bear fruit,⁸ there is in the interim a strong argument in favour of some form of network management. Proponents of net neutrality accept this truth and are very clear that net neutrality does not mean a network that is not managed,⁹ in fact as Tim Wu recognizes 'the goal of bandwidth management is, at a general level, aligned with network neutrality. Certain classes of applications will never function properly unless bandwidth and quality of service are guaranteed. Hence, the absence of bandwidth management can interfere with application development and competition.'¹⁰

p. 39

↵ Proponents of mandating net neutrality argue that there must be limits to acceptable network management for quality-of-service purposes, normally at the point where the actions of the ISP become discriminatory.¹¹ The main risk appears to be when network management techniques are applied without the knowledge of the customer. To be clear net neutrality is not about pricing: it is perfectly acceptable for an ISP to charge £20 per month for fibre at 38Mbps, £30 per month for superfast fibre at 76Mbps and £60 per month for full fibre at 1Gbps. Equally it is not about capping. If a provider offers a mobile internet package of 5Gb per month at £20 and another package at £25 for 125Gb per month that is acceptable. Net neutrality is simply the principle that you do not arbitrarily interfere in the transmission of data packets in an unclear and discriminatory fashion.

What sort of activities would normally be viewed as a breach of net neutrality principles? Most commonly these include blocking or filtering specific content or applications. For example, in 2007 it became apparent that one of the giant US cable companies, Comcast, were interfering with the ability of their cable modem customers to access BitTorrent services by resetting services that used BitTorrent packets. They were doing this as a traffic management tool to prevent BitTorrent using up available bandwidth to the detriment of other customers. Two public advocacy groups, Free Press and Public Knowledge, referred them to the Federal Communications Commission (FCC). The complaint stated that Comcast's actions violated the FCC Internet Policy Statement, particularly violating the statement's principle that 'consumers are entitled to access the lawful internet content of their choice [and] to run applications and use services of their choice'. Comcast defended its interference as necessary to manage scarce network capacity.

In August 2008 the FCC issued the results of its investigation. They found that Comcast's bandwidth management methods contravened federal policy by 'significantly impeding consumers' ability to access the content and use the applications of their choice'.¹² By the time the order was issued though Comcast had adopted new management methods and, as a result, the order effectively only required Comcast to disclose the details of those new methods and their implementation. Comcast agreed to comply with the order but also filed for review in the District of Columbia Circuit of the US Court of Appeals, claiming (among other things) that the FCC did not have jurisdiction over its network management methods. In April 2010 the court vacated the FCC's order, holding that the FCC had no authority over Comcast's internet service because 'the Commission had failed to tie its assertion of ancillary authority over Comcast's Internet service to any "statutorily mandated responsibility"'.¹³ In essence the FCC had been found to have acted *ultra vires* as they had no mandate or authority to interfere into network management capability as such interference was not ancillary to their primary statutory role. This decision was to have a major impact on the ability of US regulators to police net neutrality, as we shall see at 3.3 below.

p. 40 A second form of discrimination is access tiering, more commonly known as 'fast lanes'. This occurs when the broadband provider allows certain preferred partners ↵ access to a reserved area of the network taking their content away from the congested network. It is a little like paying for a toll road to avoid the congested public road. An example of such an agreement was revealed in 2014 when Netflix and Comcast signed a peering agreement. The agreement, it was reported, ensured 'faster and more reliable access to Comcast's broadband customers' in return for an undisclosed payment believed to be in the region of \$15m to \$20m per annum.¹⁴ Netflix later challenged the agreement as unlawful under net neutrality rules introduced by the FCC, and discussed at 3.3, however before the FCC could look at the deal changes in Federal Net Neutrality rules superseded events.

A third form of discrimination is the other side of the coin from 'fast lanes'. Throttling is the intentional slowing (or sometimes speeding) of specific content as a network management tool. Unlike access tiering this is a reactive measure employed to respond to network congestion. Like blocking and access tiering there is debate around the role of throttling with it both being a legitimate form of network management and also a threat to the principle of net neutrality that all packets may be moved equally. Throttling was reviewed by the FCC as part of their inquiry in the *Free Press & Public Knowledge* case. They found that it, like the resetting of connections, was an interference of the broadband service.¹⁵

p. 41

In 2018 the practice of throttling received some extremely bad publicity. The Mozilla Corporation and the Attorneys General of 22 US States, as well as the District of Columbia, brought a legal challenge to the FCC's *Restoring Internet Freedom Order*,¹⁶ in the United States Court of Appeals for the District of Columbia Circuit.¹⁷ In a brief filed in August 2018¹⁸ the petitioners outlined that while fighting the Mendocino Complex Fire, which started in July 2018 and which was described as the largest fire in California State history, the Santa Clara County Fire Department had their mobile data service throttled by Verizon. The briefing records that management and co-ordination of the firefighting operation was under the control of a specialist unit, OES Incident Support Unit 5262. As the brief explained 'OES 5262 relies heavily on the use of specialized software and Google Sheets to do near-real-time resource tracking through the use of cloud computing over the internet. In doing so, the unit typically exchanges 5–10 gigabytes of data per day via the internet using a mobile router and wireless connection.'¹⁹ The brief went on to state that 'in the midst of our response to the Fire, County Fire discovered the data connection for OES 5262 was being throttled by Verizon, and data rates had been reduced to 1/200, or less, than the previous speeds. Staff communicated directly with Verizon via email about the throttling, requesting it be immediately lifted for public safety purposes. Verizon representatives confirmed the throttling, but, rather than restoring us to an essential data transfer speed, they indicated that County Fire would have to switch to a new data plan at more than twice the cost, and they would only remove throttling after we contacted the Department that handles billing and switched to the new data plan.'²⁰ The brief then made a strong accusation against Verizon: 'County Fire believes it is likely that Verizon will continue to use the exigent nature of public safety emergencies and catastrophic events to coerce public agencies into higher cost plans ultimately paying significantly more for mission critical service—even if that means risking harm to public safety during negotiations.'²¹ Verizon have not denied any of this but did issue a statement saying an error was made and that restrictions should have been immediately issued once the emergency notification was received from the Fire Department.²² Despite this net neutrality advocates seized on the event as justification to repeal the *Restoring Internet Freedom Order*.²³

A final form of discrimination is zero rating. Zero rating may be seen as a form of positive discrimination, but with negative consequences.

Highlight What is zero rating?

Zero-rating²⁴ is when an ISP applies a price of zero to the data traffic associated with a particular application or class of applications (and the data does not count towards any data cap in place on the internet access service). For example, if an internet access service does not charge a user for the data used to access a specific music streaming application or all music streaming applications, then the ISPs is zero-rating those applications.

You may have seen adverts for, or even benefitted from, zero rating. Examples include the UK mobile network EE bundling of Apple Music subscriptions (with zero rated data) to its customers and Vodafone's pass scheme that allowed unlimited video or music streaming or social media use for a fixed monthly payment. For the consumer zero rating may seem like an unalloyed good. Who is going to object to receiving free data, or to the offer of unlimited data of a type for a single monthly payment (which is of course completely optional)?

p. 42

Not everyone agrees though. In December 2017 the UK telecommunications regulator Ofcom opened an investigation into the activities of two UK mobile network operators, Three and Vodafone.²⁵ The Vodafone investigation included a specific investigation into the operation of the Vodafone passes scheme. They closed the investigation in August 2018 upon receiving written assurances from Vodafone that it had made changes to the scheme.²⁶

Why are regulators interested in these schemes if they are bringing only benefits to consumers? The answer is that zero rating has clear disbenefits as well as benefits. The most obvious is that zero rating schemes ring fence certain content suppliers while shutting out others. The EE/Apple Music scheme while positive for Apple, locked out Spotify, Amazon Music, YouTube Premium and other smaller providers. The Vodafone passes scheme favoured established suppliers. The video pass for example covered video supplied only by Netflix, YouTube, Amazon Prime Video, DisneyLife, My5, TVPlayer, and UKTV Play. Other suppliers like Sky, BBC, and BT were excluded. This can have a distortive effect on competition in the supply side, entrenching some established providers while leaving competitors out in the cold. In addition, zero-rating turns service providers into gatekeepers and transforms the internet from a permission-less environment in which anyone can develop a new app or protocol and deploy it, confident that the internet treats all traffic equally, into one in which developers effectively need to seek approval from ISPs if they want their new app or service to succeed.

3.2 Regulating net neutrality

Given the potential harms of net neutrality it is unsurprising to find arguments around its regulation. However, there is little agreement on whether there should be laws or regulations to enshrine the principle of net neutrality, and if so what form these should take.

There are some who believe that as internet access is a market it should be allowed to operate as a market without external interference from regulators. This is the position held by economists Robert Hahn and Scott Wallsten. In their paper *The Economics of Net Neutrality*²⁷ they argued, 'net neutrality is actually a friendly-sounding name for price regulation.' They believe that net neutrality rules disincentivise investment by telecommunications companies in both network hardware and innovative software and services as they are unable to recover the full costs of their investments. In the words of Hahn and Wallsten 'A mandate erodes incentives to provide broadband internet access and could prevent new applications or services from ever being developed.'

p. 43

The position Hahn and Wallsten hold is prevalent in the economics literature. In his paper *Economics of Net Neutrality: A Review*,²⁸ Gerald Faulhaber notes that 'given the level of interest in network neutrality, one could be forgiven that the internet is being violated by rapacious broadband ISPs and there is not a moment to lose in protecting its openness. Since we have had broadband ISPs in the US for over a decade, one might think that the practices of blocking, discrimination, and disadvantaging competitors would be rife, and such practices well-documented. One might think, but one would be wrong.'²⁹ He goes on to note that in making the 2010 *Open Internet Report and Order* (discussed at 3.3) the FCC noted four examples of abusive behaviour in practice. This leads him to comment: 'in over a decade, there were only four examples of purported misconduct (one which was denied by the courts and another which didn't even rise to the level of a

complaint) for the entire broadband ISP industry. By any standard, four complaints about an entire industry in over a decade would seem to be cause for a commendation, not for restrictive regulations.³⁰ The argument made by Faulhaber is the standard argument from economics. Allow markets to function and then regulate for market failures only where there is clear evidence of failure or harm. However, for Faulhaber there is no evidence of such a failure or harm.³¹

Economics is a (mostly) quantitative subject and however one looks at the data it seems to suggest the market works without the need for further regulatory intervention. In their paper *Net Neutrality and Consumer Welfare*,³² Gary Becker, Dennis Carlton, and Hal Sider present a number of arguments to show that the broadband market is functioning and not in need of regulation. They state that broadband access speeds are increasing and price per MB is dropping; that customers are willing to switch providers for a better deal and that there is no evidence to support the assertion that providers will charge for access in a manner that is discriminatory or harmful. They conclude 'the FCC's net neutrality rules are motivated by the concern that broadband access providers will harm competition by disadvantaging rival content providers. This concern does not justify the imposition of net neutrality rules today due, in part, to the existence of competition in the provision of broadband access service and new competition now emerging due to entry, expansion, and upgrades of existing broadband networks. Under these circumstances, broadband access providers have strong incentives to retain subscribers by providing services and pricing models that promote consumer welfare.'³³ In examining competing models where net neutrality is set against a system where ISPs manage their networks in a discriminatory regime where each offers a priority lane and a non-priority lane to content providers, Bourreau, Kourandi, and Valletti found that 'in the discriminatory regime, where the two ISPs offer prioritized lanes, internet traffic is managed more efficiently than in the net neutrality regime. Consequently, the average level of congestion experienced by end-users is lower under discrimination than under net neutrality.'³⁴ While the authors acknowledge that 'though welfare-enhancing, the discriminatory regime might bring forth a risk of sabotage by ISPs of content providers traffic',³⁵ they believe the best model would be to regulate for this risk, not to mandate (inefficient) network neutrality.

Of course, one cannot be sure that markets will continue to function efficiently. The answer to this problem is to employ competition law to ensure that where market inefficiencies do lead to abuse these are corrected. This has been advocated by among others former Federal Trade Commissioner Maureen Ohlhausen. In her paper *Antitrust over Net Neutrality: Why We Should Take Competition in Broadband Seriously*,³⁶ she argues that 'market forces and antitrust policy can not only protect competition in ISP-related markets, but also safeguard nonmonetary goals like free speech and openness, at least to the extent that consumers share those values.'³⁷ This is achieved through vigorous application of competition law.³⁸ These combined arguments lead to the common position of most economists and at least some competition lawyers, that internet access and carriage is no different to any other market. It is best regulated through competition in the market and when or if competition fails due to abuse of market inefficiencies then competition law can step in.

These findings seem to suggest net neutrality is more of a philosophical concern than an actual concern. This brings us back to the concerns of net neutrality proponents such as Sir Tim Berners-Lee. Why should the creator of the World Wide Web hold such strong concerns if the market works? The answer is to be found in an early intervention to the debate on net neutrality. In 2006 he wrote a blog entry, unfortunately no longer

available, called *Net Neutrality: This is Serious*. He opened the entry with 12 words which explains the failure of the market approach—‘When I invented the Web, I didn’t have to ask anyone’s permission.’ Here Berners-Lee explains what is unique about the internet. The open architecture of the network, discussed at 2.1.2, allows anyone to build anything on the network. Net neutrality protects this. Markets may be generally effective regulators but at a specific level, where gatekeepers can make decisions about a single piece of technology, markets don’t always work. This suggests that net neutrality is about more than just markets and market power.

This leads to the suggestion that net neutrality is about principles rather than economics. Lucie Audibert and Andrew Murray argue this in their paper *A Principled Approach to Network Neutrality*.³⁹ Audibert and Murray believe that ‘mandating network neutrality through regulation is crucial to the protection of fundamental human rights [as well as] to ensure fair competition and innovation.’⁴⁰ According to Audibert and Murray the open internet has become part of our everyday lives and is now a vital sphere for democratic discourse and modern living.⁴¹

Highlight The essential internet

Not only does the internet provide easy and free access to information sources, it also allows people to contribute to debates in a way not possible before: With the removal of spatial and temporal bounds, and the freedom to participate anonymously or pseudonymously, the internet facilitates town-hall type gatherings and the creation of communities that might not otherwise have formed. Allowing everyone to vote is not enough for democracy to be realized—individuals must have the opportunity to voice their opinions, put them up to challenge by others, and exchange ideas. And the less costly it is to do so in terms of time and resources, the better it is. The internet provides for that, and in addition is an indispensable vector of participation in more routine, yet still crucial, activities. One needs only think of the amount of daily transactions and personal business or social activities we carry out through the internet. Almost all dealings with our banks, telephone companies, electricity and gas providers, gyms, universities, etc., are made online. Without an internet connection, we are automatically shut off from easy, streamlined access to these essential services.

p. 45 ↶ The internet is, in the words of the authors, ‘much more than a platform to post pictures of cute cats and silly videos. It has vital democratic and cultural functions and should be considered a public good to which open and free access is a fundamental right.’⁴² The authors argue this space cannot be allowed to fall under the control of a small number of gatekeepers and in order to preserve the internet’s openness and to expand its access, regulations should be enacted to prevent ISPs from carrying out illegitimate discrimination of certain types of data: net neutrality regulations.

As noted by Maniadaki, a ‘fundamental limitation in using antitrust law to protect net neutrality lies in its very nature as a tool to protect competition: competition rules will only apply in the presence of market power, or to agreements between undertakings that have the object or effect of preventing, restricting, or distorting competition. This limits the relevance of competition rules in the EU markets for the provision of

internet access services as these are largely characterized by the presence of competition.’⁴³ This leads her to conclude that ‘to the extent that net neutrality centres around the concepts of openness, plurality and content innovation, a competition law framework which is focussed, exclusively or predominately, on the protection of an efficient competitive structure and consumer welfare can only be *part of the solution*.’⁴⁴

The principled approach negates the need to explore market power, abuse, efficiency or any of the economic arguments. According to this approach we regulate the internet to ensure neutrality not because it is subject to market failure but because it should be managed according to the open network principles which applied when it was designed and built.

3.3 The legal framework

Against this backdrop the US Federal government and the European Union have both spent time examining whether there is a need to enshrine the principle of net neutrality into law. The debate in the US began around 2003 with the publication of Tim Wu’s paper *Network Neutrality, Broadband Discrimination*⁴⁵ but it really came to the fore with the FCC’s 2008 Comcast ruling.⁴⁶ Buoyed by their initial success in regulating Comcast, the FCC decided to seek public input on a new set of draft rules that would codify and supplement existing principles to safeguard internet openness. After holding a series of reviews and public meetings the FCC adopted the *Open Internet Report and Order* (the Order) in December 2010; this established three basic open internet rules designed to preserve the free and open internet. These took effect on 20 November 2011. ↵

p. 46

Highlight FCC three basic open internet rules

1. Transparency—broadband providers must disclose information regarding their network management practices, performance, and the commercial terms of their broadband services.
2. No blocking—fixed broadband providers (such as DSL, cable modem, or fixed wireless providers) may not block lawful content, applications, services or non-harmful devices. Mobile broadband providers may not block lawful websites, or applications that compete with their voice or video telephony services.
3. No unreasonable discrimination—fixed broadband providers may not unreasonably discriminate in transmitting lawful network traffic over a consumer’s broadband internet access service. Unreasonable discrimination of network traffic could take the form of particular services or websites appearing slower or degraded in quality.

In the interim the appeal in *Comcast* had been heard. This decision suggested that any attempt to enforce the Order would be fruitless as applying *Comcast*, the FCC had no authority to intervene in network and traffic management. If this were true, the Order was merely a guideline not an order. As may therefore have been expected, the efficacy of the Order was immediately challenged by a number of telecommunications

companies including Verizon and MetroPCS.⁴⁷ All these challenges were eventually consolidated into a single review before the US Court of Appeals for the Circuit of the District of Columbia.⁴⁸ In the consolidated action the telecommunications companies argued the *Comcast* decision rendered the Order *ultra vires* and in the alternative that it interfered with their First Amendment rights.

Applying *Comcast* (among other authorities) the Court found that an earlier decision of the FCC to classify broadband providers as ‘information services’ and not ‘telecommunication services’ meant that broadband service providers were not subject to so-called common carrier regulation under Title II of the Communications Act 1934.⁴⁹ The effect of this was to render invalid the provisions of the Order on anti-discrimination and anti-blocking as ‘the Commission has failed to establish that the anti-discrimination and anti-blocking rules do not impose per se common carrier obligations.’⁵⁰ The decision to vacate the key anti-blocking and anti-discrimination provisions gutted the Order of its capacity to enshrine and protect net neutrality, leaving only the provision on transparency, but brought about quite unexpected consequences and the next round of attempts to enshrine net neutrality through regulation in the United States.

p. 47 ↵ New FCC Chairman Tom Wheeler issued a statement responding to the *Verizon* decision. He stated that the FCC would not appeal the decision, but instead would establish new rules for transparency, non-discrimination and anti-blocking, based on the decision.⁵¹ In November 2014, the White House became engaged. Although the President had no power to mandate the FCC, he leveraged political pressure and made a statement calling upon the FCC to ‘implement the strongest possible rules to protect net neutrality’ and setting out four bright line rules which he suggested ‘reflect the internet you and I use every day, and that some ISPs already observe’: no blocking, no throttling, increased transparency, and no paid prioritization.

On 26 February 2015, the FCC issued a new 2015 *Open Internet Rules and Order* (the Rules).⁵² The Rules dealt with the *Verizon* decision by reclassifying broadband internet access service as a telecommunications service under Title II of the Communications Act 1934.⁵³ The Commission justified this, not only as a response to *Verizon* but because ‘our reclassification of the broadband internet access service means that we can regulate, consistent with the Communications Act, broadband providers to the extent they are “engaged” in providing the broadband internet access service.’⁵⁴ The argument made by the Commission is that in the modern world consumers see broadband providers as being similar to telecommunications providers of old: common carriers who are responsible for carrying and delivering our internet content from point to point. While this may not be technically true (the moment our email leaves our ISPs servers anyone can be carrying it by any route) it is how broadband providers advertise themselves by promoting download (and to a lesser extent upload) speeds and network security. As far as the consumer is concerned their broadband provider is the party responsible for delivering their email and making sure they can get access to Netflix. As the Commission noted: ‘the representation to retail customers that they will be able to reach “all or substantially all Internet endpoints” necessarily includes the promise to make the interconnection arrangements necessary to allow that access. As a telecommunications service, broadband internet access service implicitly includes an assertion that the broadband provider will make just and reasonable efforts to transmit and deliver its customers’ traffic to and from “all or substantially all Internet endpoints” under sections 201 and 202 of the Act. Thus, disputes involving a provider of broadband internet access service regarding internet traffic exchange arrangements that interfere with the delivery of a broadband internet access service end user’s traffic are subject to our authority under Title II of the Act.’⁵⁵

Having secured a reason to regulate broadband providers under Title II the Rules set out a new 2015 series of bright line rules, based upon President Obama's statement. ↵

Highlight FCC four basic open internet rules (2015 version)

1. No blocking—A person engaged in the provision of broadband internet access service, insofar as such person is so engaged, shall not block lawful content, applications, services, or non-harmful devices, subject to reasonable network management
2. No throttling—A person engaged in the provision of broadband internet access service, insofar as such person is so engaged, shall not impair or degrade lawful internet traffic on the basis of internet content, application, or service, or use of a non-harmful device, subject to reasonable network management
3. No Paid prioritization—A person engaged in the provision of broadband internet access service, insofar as such person is so engaged, shall not engage in paid prioritization
4. No unreasonable interference or unreasonable disadvantage standard for internet conduct—Any person engaged in the provision of broadband internet access service, insofar as such person is so engaged, shall not unreasonably interfere with or unreasonably disadvantage (i) end users' ability to select, access, and use broadband internet access service or the lawful internet content, applications, services, or devices of their choice, or (ii) edge providers' ability to make lawful content, applications, services, or devices available to end users. Reasonable network management shall not be considered a violation of this rule.

In addition to the four basic open internet rules found in the 2015 Rules it should be remembered that the transparency provision of the 2010 Order remained in effect giving five basic open internet rules in total. The rules took effect on 12 June 2015 but as may be expected before they took effect they were challenged by broadband providers.

Once promulgated the Rules were challenged by the United States Telecom Association (USTA). The case was heard in December 2015 and in June 2016 US Court of Appeals for the District of Columbia issued its decision. Somewhat surprisingly, given predictions to the contrary,⁵⁶ the Court upheld the Rules and the reclassification of internet service providers as common carriers. The majority decision, given by Tatel and Srinivasan CJs, found that the Commission had not previously misclassified broadband service providers but had simply failed to classify them. By classifying providers as a telecommunications service provider the Commission had corrected that oversight and the Order was valid.⁵⁷ Chief Circuit Judge Williams dissented from this majority position on the basis that 'the Commission's justification of its switch in classification of broadband from a Title I information service to a Title II telecommunications service fails for want of reasoned decision-making' and 'to the extent that the Commission justified the switch on the basis of new policy perceptions, its explanation of the policy is watery thin and self-contradictory.'⁵⁸ By the narrowest of

p. 49 margins the DC Court of Appeals therefore ↵ found that the classification (in the views of the majority not a reclassification) of broadband service providers as telecommunications service providers was justified and therefore that the Rules were legal.

As might be expected the petitioners appealed and sought a fresh *en banc* hearing. This was denied in May 2017 with the Court finding that ISPs were more similar to telecommunications companies in that they did not exercise editorial controls than to entertainment and broadcast companies who did.⁵⁹ There were some very strong dissenting opinions issued as the court seemed to fracture along political lines with strong words from in particular Brown CJ who claimed that ‘The President’s intervention did not result from a “failure of Congress to legislate” on the issue of Internet access regulation, but because he desired “a different and inconsistent way of his own” respecting that regulation.’⁶⁰ Whatever the strong dissenting views the decision on whether to hold a new hearing was probably by this point moot, for as the majority recognized the FCC was about to replace the Rules.

The pace of political change moves much faster than legal change. In November 2016 President Trump won election to the White House and one of his campaign pledges was to repeal the Rules. Upon taking over the reins of government in January 2017 he appointed Ajit Pai, a staunch supporter of less intervention, to be FCC Commissioner. Commissioner Pai immediately set about the task of repealing the Rules and in June 2018 the FCC implemented the *Restoring Internet Freedom Order* (the 2018 Order). This repeals the Rules. The Order also undoes the reclassification of internet broadband service providers carried out in 2015 rendering the *USTA* challenge moot. It also repealed all four open internet rules introduced in 2015, leaving only an enhanced transparency provision in place. In the words of Commissioner Pai ‘It replaces unnecessary, heavy-handed regulations that were developed way back in 1934 with strong consumer protections, increased transparency, and common-sense rules that will promote investment and broadband deployment.’⁶¹

Highlight FCC Restoring Internet Freedom Order 2018

The FCC’s new framework for protecting internet freedom has three key parts:

1. Consumer Protection—The Federal Trade Commission will police and take action against internet service providers for anticompetitive acts or unfair and deceptive practices.
2. Transparency—the FCC imposes enhanced transparency requirements. Internet service providers must publicly disclose information regarding their network management practices, performance, and commercial terms of service. These disclosures must be made via a publicly available, easily accessible company website or through the FCC’s website.
3. Removes Unnecessary Regulations to Promote Broadband Investment—the order removes the reclassification of broadband internet access service as a telecommunications service which occurred in 2015, reclassifying them once again as information service providers.

p. 50 ↪ The passing of the 2018 Order reflects the highly politicized nature of net neutrality regulation in the United States. While the FCC under the Obama White House favoured intervention and net neutrality provisions, the FCC of the Trump administration was, as might be expected, playing to free market principles. This instigated a series of developments. As mentioned at 3.1, the Mozilla Corporation and the Attorneys General of 22 US States, as well as the District of Columbia, brought a legal challenge to the 2018 Order in the United States Court of Appeals for the District of Columbia Circuit, and in October 2019 the court found in favour of the FCC.⁶²

They found that they were bound by the Supreme Court decision in *National Cable & Telecommunications Association v Brand X Internet Services*,⁶³ which found that ‘the FCC could permissibly choose not to classify cable modem service as a “telecommunications service”’.⁶⁴ It should be noted however that while agreeing they were bound by *Brand X*, one Judge, Millet CJ, reported that she was ‘deeply concerned that the result is unhinged from the realities of modern broadband service.’⁶⁵ Why was Judge Millet so concerned? As she explains ‘Brand X was decided almost fifteen years ago, during the bygone era of iPods, AOL, and Razr flip phones. The market for broadband access has changed dramatically in the interim.’⁶⁶ What difference does this make? As she explains ‘Brand X faced a “walled garden” reality, in which broadband was valued not merely as a means to access third-party content, but also for its bundling of then-nascent information services like private email, user newsgroups, and personal web page development. Today, none of those add-ons occupy the significance that they used to. Now it is impossible to deny [the] dominance of [third-party content] in the broadband experience.’⁶⁷ In short Millet CJ is suggesting the Supreme Court precedent in *Brand X*, which was about breaking down internet walled gardens of the early 2000s has been overtaken by the modern market for internet services and access. However, *Brand X* remains the law and only the Supreme Court or Congress can change that.

One small victory for supporters of net neutrality regulation was that the court found that attempts by the FCC to pre-emptively prevent States from enacting their own net neutrality laws which conflict with the 2018 Order was *ultra vires*.⁶⁸ This is important as 30 states and the District of Columbia have passed or proposed local ordinances or Acts which protect net neutrality. Prime among them is the California Internet Consumer Protection and Net Neutrality Act of 2018 which replicates and, in some cases goes further than, the 2015 Rules. Upon the passing of this Act the US Department of Justice raised an action against California stating that ‘states have no ↪ jurisdiction over the Internet’. The suit was stayed pending the outcome of *Mozilla* and then in 2021 was dropped as the new Biden administration entered the White House.⁶⁹

p. 51

This again, demonstrates the political nature of net neutrality laws in the United States. There is an ideological divide between the Democratic Party which favours intervention and the Republican Party which favours markets. With, currently, a Democratic incumbent in the White House we might expect a more interventionist approach. At the time of writing the latest development is Executive Order 14036 *Promoting Competition in the American Economy*.⁷⁰ In this President Biden encourages ‘the Chair of the Federal Communications Commission to work with the rest of the Commission, as appropriate and consistent with applicable law, to consider adopting through appropriate rulemaking ‘Net Neutrality’ rules similar to those previously adopted in ‘Protecting and Promoting the Open Internet’. This is a clear direction from the White

House to re-instate the 2015 Open Internet Order. Currently the FCC is awaiting the progress of the *Net Neutrality and Broadband Justice Bill* through Congress, but this is likely to be derailed by the 2022 mid-term elections. What will follow is hard to predict.

Net neutrality is not only an American challenge. It is equally economically important, although historically less politicized, in Europe. One of the reasons the issue was less political was a more competitive European market for internet access. In the US, fixed-line broadband access is most commonly achieved via a cable provider. This means that for many subscribers they have a limited choice of perhaps only two or three (or even one) internet access providers. In the EU most people got, and still get, their fixed-line access over digital subscriber lines or DSL (more commonly known as telephone lines). This means that the average European consumer has a choice of several access providers. In the UK for the comparison site ISP Review lists 49 competing fixed-line service providers,⁷¹ although admittedly most home users get their home broadband access from the 'big four' providers: BT, Sky, Virgin Media, and TalkTalk. The end-user can change their ISP simply by requesting their new provider to change the service over to them.⁷² Until recently the prevailing theory within Europe was that with greater competition in the internet access market, and with the regulatory authority ready to intervene should one of the behemoths of the internet access market decide to interfere with the quality of service of its customers, there was no need for proscriptive regulatory intervention.

p. 52

More recently, however, Europe's reliance on market regulation has seemed less secure. As we moved from traditional DSL lines to fibre-optic access the market ↵ narrowed. As a result, European nations moved to take steps to secure net neutrality. In April 2011 a communication from the EU Commission to Parliament and the Council entitled *The Open Internet and Net Neutrality in Europe*,⁷³ noted that despite Art.8(4)(g) of the Framework Directive⁷⁴ requiring national regulatory authorities to promote the interests of the citizens of the European Union by promoting the ability of end-users to access and distribute information or run applications and services of their choice, concerns had been raised about throttling of peer-to-peer (P2P) file-sharing or video streaming by certain providers in France, Greece, Hungary, Lithuania, Poland, and the United Kingdom and blocking or charging extra for the provision of voice over internet protocol (VoIP) services in mobile networks by certain mobile operators in Austria, Germany, Italy, the Netherlands, Portugal, and Romania.⁷⁵ The Commission noted that the EU remained committed to 'preserving the open and neutral character of the internet, taking full account of the will of the co-legislators now to enshrine net neutrality as a policy objective and regulatory principle to be promoted by national regulatory authorities'.⁷⁶ The Commission also noted though that amendments made in the 2009 Telecoms Reform Package were still being implemented by Member States and so recommended no immediate action be taken, rather they would monitor the situation.

The monitoring period ended in summer 2012. A study by the Body of European Regulators of European Communications (BEREC) found that 20 per cent of all internet users, and potentially up to half of EU mobile broadband users, had contracts that allowed their ISP to restrict services like VoIP or P2P. They further found that those fixed and mobile operators with contractual restrictions on P2P, 96 per cent of fixed-line providers, and 88 per cent of mobile providers, enforced them technically.⁷⁷ As a result, the Commission launched a public consultation into transparency, switching, and internet traffic management with an aim to preserve net neutrality. The public consultation stage closed in October 2012 after which the Commission put

together a series of packages on net neutrality and mobile roaming which led in September 2013 to the publication of the Connected Continent legislation package. Key among this was the proposal for a Regulation laying down measures concerning the European single market for electronic communications and to achieve a connected continent.⁷⁸

The Open Internet Access Regulation came into force on 30 April 2016.⁷⁹ The key net neutrality provisions are found in Arts.3–5 and allied regulations. Art.3(1) ensures open standards, while Arts.3(2) and (3) guard against discriminatory traffic management such as throttling or access tiering. ↵

p. 53

Highlight Net neutrality in Europe (Art.3)

1. End-users shall have the right to access and distribute information and content, use and provide applications and services, and use terminal equipment of their choice, irrespective of the end-user's or provider's location or the location, origin or destination of the information, content, application or service, via their internet access service.
2. Agreements between providers of internet access services and end-users on commercial and technical conditions and the characteristics of internet access services such as price, data volumes or speed, and any commercial practices conducted by providers of internet access services, shall not limit the exercise of the rights of end-users laid down in paragraph 1.
3. Providers of internet access services shall treat all traffic equally, when providing internet access services, without discrimination, restriction or interference, and irrespective of the sender and receiver, the content accessed or distributed, the applications or services used or provided, or the terminal equipment used.

The first subparagraph shall not prevent providers of internet access services from implementing reasonable traffic management measures. In order to be deemed to be reasonable, such measures shall be transparent, non-discriminatory and proportionate, and shall not be based on commercial considerations but on objectively different technical quality of service requirements of specific categories of traffic. Such measures shall not monitor the specific content and shall not be maintained for longer than necessary.

Article 3 therefore prohibits blocking, filtering, access-tiering and throttling except in cases of 'reasonable traffic management'. To ensure compliance with Art.3 transparency requirements are found in Art.4. This provides that 'providers of internet access services shall ensure that any contract which includes internet access services specifies at least the following: (a) information on how traffic management measures applied by that provider could impact on the quality of the internet access services, on the privacy of end-users and on the protection of their personal data and (b) a clear and comprehensible explanation as to how any volume limitation, speed and other quality of service parameters may in practice have an impact on internet access

services, and in particular on the use of content, applications and services.’ Both Arts.3 and 4 are backed up by the requirement that national regulatory authorities monitor service providers for compliance with the Regulation and the requirement of an annual report to BEREC.⁸⁰

p. 54 The more astute reader will have noticed that none of this appears to specifically prohibit zero rating. Even BEREC when asked the question ‘is zero-rating allowed under the Regulation?’ answered: ‘It depends’.⁸¹ The later BEREC guidelines advised that zero rating is allowed (subject to competition assessments) unless ‘all applications are blocked ↵ (or slowed down) once the data cap is reached except for the zero-rated applications [as this] would infringe Article 3(3).’⁸²

This has been litigated in two recent CJEU cases. The first, *Telenor Magyarország Zrt. v Nemzeti Média-és Hírközlési Hatóság Elnöke*,⁸³ is a 2020 decision in response to a reference from Hungary. Telenor is a Hungarian mobile network operator (MNO). It offered two specific data bundling packages ‘MyChat’ and ‘MyMusic.’ MyChat was a bundled product containing a 1GB of data for use without restriction, and which permitted the use of six specific online communication applications, Facebook, Facebook Messenger, Instagram, Twitter, Viber and Whatsapp, which were zero rated. Once the subscriber reached the plan’s data limit, they could continue to use those applications without restriction, while measures slowing down data traffic would be applied to all other applications and services. ‘MyMusic’ packages operated in the same way, although the favoured apps and services for which the applicable data limits did not apply were streaming music and/or radio services. Like ‘MyChat’, once the overall data limit was reached, the user would be able to access those preferred services without restriction while measures blocking or slowing down data traffic were applied to all other services and applications.⁸⁴

The case turned largely on the interplay between the three subsections of Art.3. Art.3(1) which establishes the affirmative right of end-users to access and distribute information and content online and to use online applications and services equally and without discrimination; Art.3(2) which allows agreements between providers and end-users as to price, data volume, data speed, and any commercial practices of the provider; and Art.3(3) which imposes affirmative obligations on the service providers to treat all internet traffic equally and without discrimination.

p. 55 The court found that Art.3(2) indicated that Art.3(1) was not intended to limit agreements and commercial practices of a given provider of internet access services to a particular end-user. In considering Art.3(1) courts should not look at such agreements individually, but provide for an overall assessment of that provider’s agreements and commercial practices.⁸⁵ This meant that practice of zero-rating was possible under Art.3 but given that ‘such [zero-rating] agreements on a significant part of the market is liable to limit the exercise of end users’ rights’⁸⁶ they could only be possible if they complied with Art.3(3). This requires that traffic management is not applied on ‘commercial considerations’ and that when Telenor applied restriction blocking or slowing down data traffic to all other services and applications but not to the preferred services they were doing so on ‘commercial considerations’.⁸⁷ As a result, Telenor were found to be in breach of Art. 3(2) as once the 1GB of data was ↵ consumed Telenor treated different sources of data differently and in a way which limited the exercise of the rights of end-users laid down in Art.3(1); and in breach of Art.3(3) as they used measures for the blocking or slowing down of traffic based on commercial considerations.⁸⁸

This left zero rating practices in limbo. Technically following *Telenor* there was nothing to prevent a provider of internet access services from adopting zero rating agreements. They could decide to offer a service unrated to end-users within Art.3. However, once the user consumes their allotted data allowance then according to *Telenor* the provider cannot distinguish between zero-rated services and other services in applying use restrictions. However, more determinative guidance was to follow in *Vodafone GmbH v Bundesrepublik Deutschland*.⁸⁹

The facts of *Vodafone* are similar to *Telenor*. Vodafone's German operation offered its customers, the option of a number of subscriptions to free 'zero tariff' options called 'Vodafone Pass' ('Video Pass', 'Music Pass', 'Chat Pass', and 'Social Pass'). These passes were all zero rated for partner services such as Apple Music or YouTube Premium. However, unlike *Telenor*, the reduction in the transfer speed provided for when the data allowance included in the basic package was used up applied also to the zero-rated services. The issue was that when the user left Germany and used data roaming the zero rated services were no longer zero rated (i.e. they consumed the data allowance of the user). The German courts believed this was a breach of the data roaming provisions of the Data Roaming Regulation (as amended).⁹⁰ The court considered not only the roaming issue but more widely the issue of zero rating observing that the roaming question assumed that a zero-rating tariff 'would itself be compatible with EU law, in particular Article 3 of Regulation 2015/2120'.⁹¹ The court found this not to be the case ruling that: 'A "zero tariff" option draws a distinction within internet traffic, on the basis of commercial considerations, by not counting towards the basic package traffic to partner applications. Consequently, such a commercial practice does not satisfy the general obligation of equal treatment of traffic, without discrimination or interference, laid down in the first subparagraph of Article 3(3) of Regulation 2015/2120'.⁹² Further, the court found it did not matter that after the data cap had been used up the two types of data were treated equally,⁹³ instead finding that all aspects of the zero rating agreement were 'contrary to the obligations arising from Article 3(3) of Regulation 2015/2120'.⁹⁴

There is now clear guidance that zero rating is incompatible with Art.3 and it is unlikely that any acceptable commercial form of zero rating can take place. Non-commercial zero rating might be allowed though.

p. 56

Although Art.3(3) does not distinguish between commercial and non-commercial services there are several recent examples of zero rating of public services. In 2020, UK MNOs agreed to zero rate data used for the NHS website and the NHS COVID-19 tracing app, so that customers could use these services without it affecting their data usage limits.⁹⁵ In 2021 UK MNOs, agreed to zero rate a number of websites supporting victims of crime,⁹⁶ while recently a number of MNOs have zero rated a number of advisory sites including Citizens Advice, the Money Advice Service and the debt charity, Step Change in response to the cost of living crisis.⁹⁷ Although technically a breach of Art.3 it is unlikely such altruism will lead to enforcement actions.

Finally, in a theme which will develop throughout this book we need to ask what changes Brexit has made to this settlement. In the short term, there is little impact. The UK has adopted and domesticated the provisions of the Regulation by The Open Internet Access (Amendment etc.) (EU Exit) Regulations 2018.⁹⁸ The main amendments made are to remove the role of BEREC and to designate Ofcom as the regulatory authority. The main operative provisions of Arts.3–5 are adopted into domestic law. Following Brexit Ofcom launched a net neutrality review⁹⁹ which received a number of responses from leading telecommunication companies such as BT (who bullishly called for wholesale amendments to the current framework) and content providers such as Netflix, most of whom supported the status quo. In October 2022, this was taken forward in the form of a

Consultation which seems to suggest the UK may take a more liberal approach to zero rating than the EU post *Vodafone*, in particular by explicitly allowing for zero-rating of non-commercial content.¹⁰⁰ The review will close in January 2023 after which we might expect the next steps to be published.

3.4 Conclusions

p. 57

Clearly the issue of net neutrality will remain to the fore of policy debate both in the United States and Europe given the importance of the principle in both policy and economic terms. At the moment in the United States Net Neutrality is as much a political issue as it is a policy/legal issue. The current administration are possibly about to push back on the actions of the previous administration by having the FCC overturn the 2018 Order and moving back to the standards of the 2015 Rules. With both chambers of the United States Congress finely balanced following the 2022 mid-term elections it seems the debate between market regulation and legal intervention may not be settled there for some time.

Here in Europe the position is (a little) clearer. The Open Internet Access Regulation clearly mandates net neutrality in all areas, including following the decision in *Vodafone*, zero rating. However, even here political complexities remain. For example, I have cited *Telenor* and *Vodafone* as authority for the position that zero rating is a breach of Art.3(3) of the Open Internet Regulation. Technically, this is not the case in the UK. As both decisions were issued after Brexit day neither are authority in UK Law. UK courts and tribunals are not bound by CJEU decisions made after 31 December 2020 (except those which were referred from the UK pre-exit). That does not mean the UK cannot look to Europe for guidance. Under the terms of the European Union (Withdrawal) Act 2018, all domestic courts may have regard to post-exit CJEU judgments when construing retained EU law (s.6(2)). Thus, the decisions in *Telenor* and *Vodafone* are influential at best and may be ignored by UK courts and tribunals. Clearly, the politicized nature of the net neutrality debate will ensure continued developments.

Test Questions

Question 1

Should regulators be intervening in the free and open market for internet access and transfers of data by implementing specific 'net neutrality' regulations and orders? Is the preservation of a free and open internet a self-standing regulatory objective?

Question 2

At present, it is argued, the existence of market failures requires government regulation to ensure net neutrality. What are these market failures?

Question 3

What aspects of internet architecture should be regulated applying Net Neutrality principles? Should zero rating for instance be allowed or disallowed applying Net Neutrality principles? What about ‘fast’ and ‘slow’ lanes? Should throttling and blocking be allowed or disallowed?

Further Reading

Books

L Belli and P De Filippi (eds.), *Net Neutrality Compendium* (Springer 2015)

P Maillé and B Tuffin, *From Net Neutrality to ICT Neutrality* (Springer 2022)

C Marsden, *Network Neutrality* (Manchester UP 2017)

V Pickard and D Berman, *After Net Neutrality* (Yale UP 2019)

p. 58 Chapters and articles

L Audibert and A Murray, ‘A Principled Approach to Network Neutrality’ (2016) 13 *Script-Ed* 118

AJ Carrillo, ‘Having your Cake and Eating it too? Zero-rating, Net neutrality, and International Law’ (2016) 19 *Stanford Technology Law Review* 364

A Cooper and I Brown, ‘Net Neutrality: Discrimination, Competition, and Innovation in the UK and US’, (2015) 15 *ACM Transactions on Internet Technology* 2 (2015)

C Hooten, ‘Testing the Economics of the Net Neutrality Debate’ (June 2020) 44 *Telecommunications Policy* 101869

C Marsden, ‘Comparative Case Studies in Implementing Net Neutrality: A Critical Analysis of Zero Rating’ (2016) 13 *SCRIPTed* 1

T Wu, ‘Network Neutrality, Broadband Discrimination’ (2003) 2 *Journal of Telecommunications and High Technology Law* 141

Notes

¹ T Berners-Lee, ‘In defense of Net Neutrality’, *Wall Street Journal*, 22 June 2017.

² Berners-Lee, *ibid*; Federal Communications Commission, *Restoring Internet Freedom* 83 FR 7852 2018 [WC Docket No. 17–108; FCC 17–166]; President J Biden *Executive Order 14036 on Promoting Competition in the American Economy*, 9 July 2021.

³ High Tech Broadband Coalition, *Appropriate Framework for Broadband Access to the Internet over Wireline Facilities* (CC Docket No. 96–45 (2002)); High Tech Broadband Coalition, *Appropriate Regulatory Treatment for Broadband Access to the Internet over Cable Facilities* (CC Docket No. 96–45 (2002)). C Marsden, *Network Neutrality: From Policy to Law to Regulation* (Manchester University Press 2017).

⁴ In 2021, Video traffic dominated the demand for bandwidth and accounted for 53.72 per cent of total internet traffic, with two sites, Netflix and YouTube, accounting for 24 per cent of global traffic. In American homes these two sites accounted for 34.87 per cent of traffic while in Europe and the Middle East they accounted for 28.85 per cent. Sandvine, *Global Internet Phenomena Report 2022* <https://www.sandvine.com/global-internet-phenomena-report-2022> <<https://www.sandvine.com/global-internet-phenomena-report-2022>>.

⁵ A Cooper and I Brown, 'Net Neutrality: Discrimination, Competition, and Innovation in the UK and US' (2015) 15 *ACM Transactions on Internet Technology* 2; T Wu, 'Network Neutrality, Broadband Discrimination' (2003) 2 *Journal of Telecommunications and High Technology Law* 141.

⁶ L Lessig, *The Future of Ideas* (Random House 2001), ch.10; R Frieden, 'Can Internet Service Providers Lawfully Enhance, or Degrade Third Party Content?' (18 May 2017) SSRN <http://dx.doi.org/10.2139/ssrn.2970716> <<http://dx.doi.org/10.2139/ssrn.2970716>>.

⁷ Wu (n.5) found that among the activities restricted were: Any commercial or business use of facilities; operating a server; overusing bandwidth; and in some cases even setting up a home network or wireless network (160–6).

⁸ The UK government has a gigabit broadband target. It was set out in the Conservative Manifesto in 2019 that they would deliver 'full fibre and gigabit-capable broadband to every home and business across the UK by 2025'. This was later revised to a target of 85 per cent in the 2020 *National Infrastructure Strategy*, before being amended again to 'as close to 100% as possible' by 2030 in the 2022 *Levelling Up White Paper*. Meanwhile the industry website *Think Broadband* reported that only 70.42 per cent of UK of premises had gigabit-broadband available in the week commencing 5 September 2022.

⁹ Wu (n.5); RS Lee and T Wu, 'Subsidizing Creativity through Network Design: Zero-Pricing and Net Neutrality' (2009) 23 *Journal of Economic Perspectives* 61; Marsden (n.3) Ch.1.

¹⁰ n.5, 155.

¹¹ Ibid, 142ff.

¹² *In re Formal Complaint of Free Press & Public Knowledge Against Comcast Corporation for Secretly Degrading Peer-to-Peer Applications*, 23 FCCR 13,028 at 13,054 (2008).

¹³ *Comcast Corp. v FCC*, 600 F 3d 642, 661 (2010).

¹⁴ D Crow, 'Having Signed the Multiyear Contract, Netflix Wants To Put Comcast Genie Back in "Fast Lane" Bottle', *Financial Times*, 9 November 2014.

¹⁵ n.12.

¹⁶ FCC-17–166 (14 December 2017). Discussed in depth at 3.3.

¹⁷ *Mozilla Corporation v FCC, et al.* 0:18-rev-01051.

¹⁸ From: https://ag.ny.gov/sites/default/files/nn_govt_petitioners_brief_and_addendum:final_filed.pdf <https://ag.ny.gov/sites/default/files/nn_govt_petitioners_brief_and_addendum:final_filed.pdf>.

¹⁹ Ibid, [6], [8].

²⁰ Ibid, [9].

²¹ Ibid, [11].

²² C Lecher, 'Verizon Throttled California Fire Department During Wildfire Crisis', *The Verge*, 21 August 2018.

²³ G Sohn, 'Verizon Couldn't Have Restricted Santa Clara County's Internet Service During the Fires Under Net Neutrality', *NBC News*, 24 August 2018; J Kruzel, 'Could Net Neutrality Have Shielded California Firefighters from Throttling?', *Politifact*, 7 September 2018.

²⁴ BEREC, *What is Zero Rating?* <https://www.berec.europa.eu/en/what-is-zero-rating> <<https://www.berec.europa.eu/en/what-is-zero-rating>>.

²⁵ https://www.ofcom.org.uk/about-ofcom/latest/bulletins/competition-bulletins/open-cases/cw_01210 <https://www.ofcom.org.uk/about-ofcom/latest/bulletins/competition-bulletins/open-cases/cw_01210>.

²⁶ https://www.ofcom.org.uk/about-ofcom/latest/bulletins/competition-bulletins/all-closed-cases/cw_01219 <https://www.ofcom.org.uk/about-ofcom/latest/bulletins/competition-bulletins/all-closed-cases/cw_01219>. Vodafone passes were discontinued in January 2019. EE Smart Benefits, including the Apple Music benefit, are no longer zero rated.

²⁷ (2005) 3 *The Economist's Voice*: <http://www.bepress.com/ev/vol3/iss6/art8/> <<http://www.bepress.com/ev/vol3/iss6/art8/>>.

²⁸ (2011) 3 *Communications & Convergence Review* 53.

²⁹ Ibid, 56.

³⁰ Ibid, 57.

³¹ Ibid, 58.

³² (2010) 6 *Journal of Competition Law & Economics* 497.

³³ Ibid, 519.

³⁴ M Bourreau, F Kourandi, and T Valletti, 'Net Neutrality with Competing Internet Platforms' (2015) 63(1) *The Journal of Industrial Economics* 30, 68.

³⁵ Ibid.

³⁶ (2016) 15(1) *Colorado Technology Law Journal* 119.

³⁷ Ibid, 133.

³⁸ Ibid, 141.

³⁹ (2016) 13 *Script-Ed* 118.

⁴⁰ Ibid, 120.

⁴¹ Ibid, 134.

⁴² Ibid, 119.

3. Net neutrality

⁴³ K Maniadaki, 'Net Neutrality Regulation in the EU: Competition and Beyond' (2019) 10(7) *Journal of European Competition Law & Practice* 479, 482.

⁴⁴ *Ibid*, 488.

⁴⁵ n.5.

⁴⁶ n.12.

⁴⁷ *Verizon v FCC*, Case No. 11–1014 (D.C. Cir. 20 January 2011); *MetroPCS Communications et al. v FCC*, Case No. 11–1016 (D.C. Cir. 24 January 2011).

⁴⁸ *Verizon Communications Inc. v FCC* 740 F.3d 623 (D.C. Cir. 2014).

⁴⁹ *Ibid*, 9.

⁵⁰ *Ibid*, 4.

⁵¹ FCC, *Statement by FCC Chairman Tom Wheeler on the FCC's Open Internet Rules*, 19 February 2014: <https://www.fcc.gov/document/statement-fcc-chairman-tom-wheeler-fccs-open-internet-rules> <<https://www.fcc.gov/document/statement-fcc-chairman-tom-wheeler-fccs-open-internet-rules>>.

⁵² FCC15-24: https://apps.fcc.gov/edocs_public/attachmatch/FCC-15-24A1.pdf <https://apps.fcc.gov/edocs_public/attachmatch/FCC-15-24A1.pdf>.

⁵³ *Ibid*, [59].

⁵⁴ *Ibid*, [339].

⁵⁵ *Ibid*, [204].

⁵⁶ JP Tuthill, 'FCC Throws in the Towel, but Public Has Right To Know Why', *SF Gate*, 25 February 2014 <http://www.sfgate.com/opinion/openforum/article/FCC-throws-in-the-towel-but-public-has-right-to-5267613.php> <<http://www.sfgate.com/opinion/openforum/article/FCC-throws-in-the-towel-but-public-has-right-to-5267613.php>>; A Hurst, 'Neutering Net Neutrality: What *Verizon v FCC* Means for the Future of the Internet' (2015) 7 *Hastings Science and Technology Law Journal* 43.

⁵⁷ *United States Telecom Association v FCC* 825 F.3d 674, 728 (D.C. Cir. 2016).

⁵⁸ *Ibid*, 790.

⁵⁹ *United States Telecom Association v FCC* No. 15–1063 (D.C. Cir. 2017).

⁶⁰ *Ibid*, 45.

⁶¹ <https://docs.fcc.gov/public/attachments/DOC-351481A1.pdf> <<https://docs.fcc.gov/public/attachments/DOC-351481A1.pdf>>.

⁶² *Mozilla Corporation v Federal Communications Commission and the United States of America* No. 18–1051 1 October 2019. [https://www.cadc.uscourts.gov/internet/opinions.nsf/FA43C305E2B9A35485258486004F6D0F/\\$file/18-1051-1808766.pdf](https://www.cadc.uscourts.gov/internet/opinions.nsf/FA43C305E2B9A35485258486004F6D0F/$file/18-1051-1808766.pdf) <[https://www.cadc.uscourts.gov/internet/opinions.nsf/FA43C305E2B9A35485258486004F6D0F/\\$file/18-1051-1808766.pdf](https://www.cadc.uscourts.gov/internet/opinions.nsf/FA43C305E2B9A35485258486004F6D0F/$file/18-1051-1808766.pdf)>.

⁶³ 545 U.S. 967 (2005).

⁶⁴ N.62, 16.

⁶⁵ Ibid, 147.

⁶⁶ Ibid.

⁶⁷ Ibid, 147–8.

⁶⁸ Ibid, 139.

⁶⁹ G Maddaus, 'Biden's Justice Department Drops Legal Challenge to California Net Neutrality Law', *Variety*, 8 February 2021 <https://variety.com/2021/digital/news/biden-net-neutrality-department-of-justice-1234903450/> <<https://variety.com/2021/digital/news/biden-net-neutrality-department-of-justice-1234903450/>>.

⁷⁰ 9 July 2021 <https://www.whitehouse.gov/briefing-room/presidential-actions/2021/07/09/executive-order-on-promoting-competition-in-the-american-economy/> <<https://www.whitehouse.gov/briefing-room/presidential-actions/2021/07/09/executive-order-on-promoting-competition-in-the-american-economy/>>.

⁷¹ <https://www.ispreview.co.uk/review/top50.php> <<https://www.ispreview.co.uk/review/top50.php>>.

⁷² <http://consumers.ofcom.org.uk/internet/broadband-switching/switching-broadband-provider/> <<http://consumers.ofcom.org.uk/internet/broadband-switching/switching-broadband-provider/>>.

⁷³ COM(2011) 222 final.

⁷⁴ Dir.2002/21/EC.

⁷⁵ n.73, [4.1].

⁷⁶ EU telecoms reform package [2009] OJ L337.

⁷⁷ BEREC, *A View of Traffic Management and other Practices Resulting in Restrictions to the Open Internet in Europe* (29 May 2012).

⁷⁸ COM(2013) 627 final.

⁷⁹ Reg.2015/2120.

⁸⁰ Art.5(1).

⁸¹ n.24.

⁸² BEREC, *Guidelines on the Implementation of the Open Internet Regulation* (June 2020), [41] https://www.berec.europa.eu/sites/default/files/files/document_register_store/2020/6/BoR_%2820%29_112_BEREC_Guidelines_on_the_implementation_of_the_OI_Regulation.pdf <https://www.berec.europa.eu/sites/default/files/files/document_register_store/2020/6/BoR_%2820%29_112_BEREC_Guidelines_on_the_implementation_of_the_OI_Regulation.pdf>.

⁸³ Joined cases C-807/18 and C-39/19: ECLI:EU:C:2020:708.

⁸⁴ Ibid, [10]–[11].

⁸⁵ Ibid, [42].

⁸⁶ Ibid, [46].

⁸⁷ Ibid, [51]–[52].

⁸⁸ Ibid, [55].

⁸⁹ ECLI:EU:C:2021:675.

⁹⁰ Regulation (EU) No. 531/2012. NB! This Regulation is no longer in force having been repealed by Regulation (EU) 2022/612.

⁹¹ n.89, [16].

⁹² Ibid, [28].

⁹³ Ibid, [29].

⁹⁴ Ibid, [33].

⁹⁵ <https://www.theyworkforyou.com/wrans/?id=2022-01-27.113059.h> <<https://www.theyworkforyou.com/wrans/?id=2022-01-27.113059.h>>.

⁹⁶ <https://www.gov.uk/government/news/data-charges-removed-for-websites-supporting-victims-of-crime> <<https://www.gov.uk/government/news/data-charges-removed-for-websites-supporting-victims-of-crime>>.

⁹⁷ <https://www.ispreview.co.uk/index.php/2022/07/o2-and-virgin-mobile-uk-zero-rate-mobile-data-on-more-websites.html> <<https://www.ispreview.co.uk/index.php/2022/07/o2-and-virgin-mobile-uk-zero-rate-mobile-data-on-more-websites.html>>.

⁹⁸ SI 2018/1243.

⁹⁹ <https://www.ofcom.org.uk/consultations-and-statements/category-2/call-for-evidence-net-neutrality-review> <<https://www.ofcom.org.uk/consultations-and-statements/category-2/call-for-evidence-net-neutrality-review>>.

¹⁰⁰ Ofcom, *Net Neutrality Review* (October 2022) https://www.ofcom.org.uk/__data/assets/pdf_file/0028/245926/net-neutrality-review.pdf <https://www.ofcom.org.uk/__data/assets/pdf_file/0028/245926/net-neutrality-review.pdf>.

© Oxford University Press 2023

Find This Title

In the OUP print catalogue <<https://global.oup.com/academic/product/9780192893529>>